



/Users/halpern/teaching/352/labs/manual/master/rev_2022/alpha_prelab.tex

Alpha Particles Prelab

Simulate the passage of an α -particle through air.

Our approach will be to consider a layer of air so thin that any single alpha particle might interact once, or not at all, but never twice as it passes through the thin layer. This way the behavior of each layer is simple, and we can iterate, passing alphas through a layer again and again, to simulate passing through a lot of air.

Start with a population of particles with a gaussian distribution of energies centered at E_o . Imagine a layer of air thin enough that the typical particle passes through without interaction, but some fraction, F , lose energy dE . The important point here is to choose F small enough that particles interact zero or one times, but never two in a single passage. Choose how $F(E, dE)$ depends on dE and E and simulate a single passage. Plot the resulting energy distribution.

Iterate energy loss a few times. Be sure to remove from the iterated distribution particles which have stopped. Maybe it is easiest to pick some lower bound in E , and only process particles on each iteration whose energy is above this threshold.

By choosing F to describe a thin layer of air, and then iterating the energy process again and again, you are modelling the energy loss as you travel through N layers, each of thickness captured by F . Your calculation should mimic travelling an arbitrary distance through air of arbitrary density. How does the energy distribution behave as you let N grow?